Python for computational Problem Solving · Unit_03 · Unit 3: Functions, Functional Programming and Modules

DRAFT — UE26CS151A · Unit_03 · Unit 3: Functions, Functional Programming and Modules · PIL

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

PIL

1 PIL

Study focus: PIL. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000110 · EV0000111

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we are going to explore Pillow, the modern version of Python Imaging Library, also called PIL.

Key details

- Simple and powerful library that helps us open images, edit images, and save images in multiple formats.
- It specially deals with image processing tasks.
- Pillow is widely used because it is easy to understand, support many image formats, and includes built-in functions for tasks like resizing, cropping, rotating and filtering images.
- To install Pillow module, we simply use Pip install pillow.
- To start with below, we first import the image class.
- Here's a basic code to open an image and display it on the screen.
- Image.show displays it using your system's default viewer.
- Pillow supports JPEG, PNG, BMP, JIF, TIF, and many more image formats.
- Once we load the image, we can check some important details like format, which tells the file type like JPEG, or GIF, or Tiv, etc.
- Then the size, which gives the width and height in pixels.
- Then the mode shows the colour representation such as RGB or grayscale.
- These properties help us understand the image before processing it.
- Resizing is one of the most common operations in image processing.
- Here's how we resize an image.
- But it may distort the aspect ratio.
- To maintain the aspect ratio we can use thumbnail.
- This ensures the image is resized without stretching or squashing.

- converting the image modes.
- You can use convert function to convert to an image to a different colour mode.
- The common colour modes are one, which is pure black and white, it's just one bit.
- L, which is grayscale.
- All the shades of gray colour will be represented in an 8-bit representation.
- Then RGB, red, green, blue colour, three channels, and this is also an 8-bit representation.
- Red, green, blue, and A stands for alpha, which is represents the transparency.
- Then C M Y K, which stands for Syan, Majenta, Low, and Black.
- Mostly used in printing.
- Mode conversion is especially useful in machine learning preprocessing, where special formats are required.
- To save an image that you have worked on, you can use save function.
- Then to rotate an image, you can use rotate function and give the degrees as argument.

Example or application from the lecture

- Let's begin.
- So this example is importing the image class and using the functions open and show.
- Next one, let's take a look into few more essential operations like saving images.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 05:00–09:59

To crop an image, we can use crop function and the syntaxes you can give the pixel coordinates as left, upper, right, lower to extract the specific region from the image.

Key details

- These operations make Pillow extremely useful for automation workflows, image editing tasks, and pre processing pipelines.
- To summarize, Pillow is a very powerful and big enough friendly Python library for image processing.
- We learned how to use open images, display them, resize, convert color modes, rotate, crop and save images.
- With these foundations you can build a wide variety of image based applications from simple editors to machine learning pipelines.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000114 · EV0000115 · EV0000116 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
PIL transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/015_PIL__b612657fcad391f3.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
PIL transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/015_PIL__b612657fcad391f3.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
PIL transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/015_PIL__b612657fcad391f3.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
PIL transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/015_PIL__b612657fcad391f3.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
2. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.

4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
6. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00042	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0025. Sidecar variants compare...
SRC00043	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0025. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00044	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0025. Sidecar variants compare...
SRC00045	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0025. Sidecar variants compare...

C Revision and retrieval practice